# Model Collapse on Manifolds: a Closed-Form Capacity Floor and a Data-Anchoring Fix

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**Abstract**

Generative models are increasingly retrained on data that includes their own past outputs, and for diffusion models this *self-consuming* loop is known to degrade quality from one generation to the next. We settle the case that matters most in practice and that the sharpest prior theory [1] repairs only for smooth densities and explicitly leaves open—data concentrated on a thin low-dimensional manifold—with matching negative and constructive results. *Negatively*, we prove a closed-form *capacity floor*: any truncated reverse-SDE sampler whose normal score slope is capped at $\bar{\kappa}$ retains excess width of at least $1/(8\bar{\kappa}^2)$ on every pass, with no dependence on the sampling schedule. The mechanism is geometric: near a manifold of width $\sigma$ the score a diffusion model must represent is steep only inside a short, finite window of noise levels, so a slope-capped network bakes in excess width there that later denoising cannot remove. A model held at a fixed slope ceiling therefore collapses without bound as the manifold thins, and staying bounded requires a slope budget growing like $\sqrt{J}$ in the Fisher information $J$. Throughout, the hypothesis bounds the network's normal Lipschitz norm, not its parameter count; the link from architecture size to attainable slope is measured here, not proved. Empirically the trap is two-horned: the stochastic sampler sticks $\sim 11\times$ too wide, while the deterministic one collapses to a point; neither reaches the data. *Constructively*, we prove that periodically re-matching the model's output to a small, *fixed* pool of real examples makes the loop *memoryless*, so that error stops compounding for every positive fraction of real data. The match rescales the output to the *local width* of the reference; it does not re-insert the reference examples into training. Over a 20-generation head-to-head the prior state of the art plateaus at $9.8\sigma^2$ while anchoring holds $1.01\sigma^2$, and on pixel-space MNIST anchoring removes $\sim$81% of the off-manifold collapse. A preliminary scale-up to CIFAR-10 in raw pixel space (3,072 dimensions), across three seeds, reproduces the separation and, by three independent nearest-neighbor metrics, shows the anchor keeps full coverage of the real data, ruling out mode collapse. The core experiments run on CPU and all are reproducible from released scripts.

## 1 Introduction

**In plain terms.** Train an image generator, use it to make pictures, then train the next generator partly on those pictures, and repeat. Each round the model drifts a little further from real data, the way a photocopy of a photocopy loses detail. This is *model collapse*. Recent theory shows the drift can be repaired when the data is "smooth," but leaves open the case that actually describes images, audio, and most real data: points that lie on a thin surface (a *manifold*) inside a much larger space. This paper handles that case. We prove there is a hard floor on how faithful a fixed-size model can stay, set by a single number, how sharp an edge the network can represent, that no amount of tuning removes. Then we give a cheap fix: each round, gently rescale the model's output so its fine-scale spread matches a small, fixed sample of real data. The fix stops the drift from compounding, for any positive amount of fresh real data.

Web-scale corpora already contain machine-generated data, and synthetic data is used deliberately in modern training pipelines. When a generative model is repeatedly retrained on a pool containing its own previous outputs, quality degrades across generations, a phenomenon called *model collapse* [2–5]. For diffusion models, Khelifa et al. [1] give the sharpest analysis to date: with a fraction $\lambda$ of fresh real data per generation and an annealed truncation schedule $t_0(g) \to 0$, the recursion converges back to the data distribution, *provided the data has finite Fisher information* (their Case 1). They explicitly leave open their Case 2: data supported on, or concentrated near, a low-dimensional manifold, where the Fisher information $J \asymp (d-k)/\sigma^2$ diverges as the noise width $\sigma \to 0$. Essentially all natural data (images, audio, molecular conformations) is of this second kind [6, 7].

This paper answers Case 2.

**The idea in one paragraph.** The hardest property to preserve across generations is the data's *thinness*—the exact width of a pen stroke, the sharp edge of a shape. A finite-capacity denoiser can reproduce that thinness only inside a brief window of noise levels, so a small, fixed amount of blur is baked in every generation and accumulates. This yields our two results, made precise below: a *negative* one—the blur has an exact floor set by a single number, the steepest slope the network can represent, that no training or sampling schedule lowers—and a *constructive* one—the accumulation is cancelled by periodically re-matching the model's output to the *local width* of a small, unchanging set of real examples, without copying those examples back into training.

**The negative half** is a *capacity floor* (Section 5). The score that a diffusion model must represent near a $\sigma$-thin manifold has normal slope $\kappa(t) = s(t)/\gamma^2(t)$ which is *non-monotone* in diffusion time: it peaks at $\kappa_{\max} = 1/(2\sigma)$ near $t \approx \sigma^2$ and falls back toward zero as $t \to 0$ (Figure 1a). A network whose normal slope is bounded by $\bar{\kappa}$ therefore under-contracts on a *finite band* of diffusion times; the excess variance frozen in while crossing that band can never be removed afterwards, because too few contraction e-folds remain below the band. We prove that in the singular limit the resulting floor is a universal function of the single ratio $\rho = \bar{\kappa}/\kappa_{\max}$ and admits *closed forms*: under no assumption beyond the slope bound, the floor is exactly $\sigma^2/(2\rho^2) = 1/(8\bar{\kappa}^2)$, so a fixed network collapses relatively without bound as $\sigma \to 0$, and bounded collapse requires $\bar{\kappa} = \Omega(1/\sigma) = \Omega(\sqrt{J})$. No truncation schedule enters these statements: annealing targets a term the schedule can kill, while the floor lives in a term it cannot touch. Empirically the trap is *two-horned* (Figure 2): keeping the reverse-SDE noise, the sampler sticks $\sim 11\times$ too wide; removing it, the recursion collapses to a single point. Neither branch reaches the data.

**The constructive half** is an anchoring operator (Section 6). A bidirectional, per-axis, local moment match of the training pool (and sampler output) to a small *stale* reference of real data (2,000 samples drawn once at generation zero) pins the normal variance at the reference value each generation. We prove this makes the recursion *memoryless*: the fixed-point equation loses its dependence on the previous generation, the critical fraction disappears, and the dynamics of Khelifa et al. [1]'s Theorem 4.1 are restored in the singular regime. In a 20-generation head-to-head against their annealed-truncation remedy (Figure 5), the annealed arm plateaus at $9.8\sigma^2$ while the anchored arms hold $1.01\sigma^2$, flat, on both samplers we test; on real pixels (MNIST), anchoring closes $\sim$81% of the collapse gap (Figure 6); and a preliminary CIFAR-10 pixel recursion (3,072 dimensions, three seeds) reproduces the anchored/unanchored split with full mode coverage (Figure 7).

**Contributions.**

1. **A law of collapse** (Theorem 1): under a measured one-dimensional concave response law, the variance recursion has a unique plateau for $\lambda > \lambda^* = c_\infty/(1 + c_\infty)$ and no plateau below; truncation enters only additively, which makes precise why annealing schedules cannot help singular data.
2. **A closed-form capacity floor** (Theorem 2, Corollary 1): any reverse-SDE sampler with normal score slope $\leq \bar{\kappa}$ is floored; in the singular limit the floor is $1/(8\bar{\kappa}^2)$ unconditionally and $\big((1 + 3e^{-4})/8\big)\bar{\kappa}^{-2}$ for the empirically observed no-overshoot class: two constants that differ by only 5.5%. Bounded relative collapse as $\sigma \to 0$ requires $\bar{\kappa} \geq 1/(2\sigma\sqrt{2C})$.
3. **A memorylessness theorem for anchoring** (Theorem 3): matching pool and output normal variance to a stale real reference removes the recursion's memory, for every $\lambda > 0$, with an error bound independent of the ambient dimension.
4. **Falsifiable empirics**, each targeting a named claim:
   - *Floor anatomy & intervention*: read at the ceiling the theorem constrains, the law's envelope explains 81% of the measured single-pass floor at $\sigma = 0.05$ (against 58% read under $p_t$, and with half the seed spread), the remainder resolving into one further measured channel (in-band profile sag); the margin narrows with tube width and the comparison fails at $\sigma = 0.13$, where the flat-tube idealisation expires; and across five training protocols that move the ceiling $3.5 \to 7.5$, every measured floor stays above its proven bound.
   - *Two-horned ablation*: separates the floor's existence (sampler noise) from its size (slope ceiling).
   - *Head-to-head*: a 20-generation, 5-seed win over the state-of-the-art remedy, with a replay-buffer control at the anchor's exact information budget that separates anchoring from replay, and a subcritical-fraction run exhibiting the law's no-plateau regime.
   - *Real and scaled data*: a pixel-space MNIST recursion, and a preliminary CIFAR-10 pixel recursion (three seeds) that rules out mode collapse by three independent nearest-neighbor metrics.

Throughout we keep the epistemic status of each claim explicit: *we prove* (mathematics, given stated assumptions), *we observe* (multi-seed experiments), or *we model* (measured structural laws that we do not derive). Section 8 collects everything in the third category.

## 2 Related work

**Model collapse.** Collapse under recursive training is documented and analyzed across model families [2–5, 8]; accumulation (rather than replacement) of data and sufficient fresh-data fractions mitigate it in the absolutely continuous setting. Khelifa et al. [9] bound error propagation in recursively trained diffusions via a discounted accumulation argument, but their assumptions require absolutely continuous data (structurally excluding manifolds), with fixed truncation and upper bounds rather than laws. Khelifa et al. [1] characterize the limiting collapse distribution spectrally and prove annealed truncation converges for finite-Fisher data (their Case 1), stating manifold-supported data (Case 2) as open. We give the Case 2 answer: a floor theorem for their sampler class, and an operator restoring their Case 1 dynamics.

**Diffusion on manifold data.** That diffusion models detect and concentrate on data manifolds is established [7, 10]. The singular ($\sigma \to 0$) score is the object of a growing one-shot literature: optimal denoising under low regularity [11], provable learning under the manifold hypothesis [12], and mitigation of the score singularity [13]. These works are one-generation; our subject is the *recursion*, where a per-generation floor compounds into a permanently displaced plateau. We use the Tweedie/posterior-mean jump [11, 14] as a raw-error reducer inside the loop and are explicit that it does not, by itself, break the floor (Section 6).

**Local geometry estimation.** Our anchoring operator estimates local tangent frames by $k$NN-PCA; its error analysis imports standard local-PCA perturbation rates [15, 16]. The operator needs only *shared* frames between pool and reference, not correct ones, which is why these rates enter only at second order (Lemma 4).

## 3 Setup

**Data model ($\sigma$-tube).** Let $M \subset \mathbb{R}^d$ be a compact smooth $k$-dimensional manifold with reach $\tau > 0$. Data are $X_0 = \varphi(U) + \sigma N_\perp$, with $U$ uniform on $M$ and $N_\perp$ standard normal in the $(d-k)$-dimensional normal space; $\sigma > 0$ is the tube width. The singular regime is $\sigma \to 0$; the Fisher information of the marginal scales as $J \asymp (d-k)/\sigma^2 \to \infty$ (Case 2 of 1).

**Forward process and the normal coordinate.** We use the variance-preserving OU forward process $X_t = a(t)X_0 + s(t)Z$ with $a(t) = e^{-t/2}$, $s^2(t) = 1 - e^{-t}$ [17, 18]. Fix a point of $M$ and work in a normal coordinate $x \in \mathbb{R}$, treating the fiber marginal as Gaussian and decoupled from tangential position (*flat-tube idealization*; curvature enters at $O(s^2/\tau)$ and is tracked where it matters). Then $X_0 \sim \mathcal{N}(0, \sigma^2)$, $X_t \sim \mathcal{N}(0, \gamma^2(t))$ with $\gamma^2 = a^2\sigma^2 + s^2$, and the denoising target is linear:

$$\varepsilon^*(x,t) = \kappa(t)\, x, \qquad \kappa(t) = \frac{s(t)}{\gamma^2(t)}, \qquad \nabla \log p_t(x) = -\frac{x}{\gamma^2(t)}. \tag{1}$$

$\kappa$ is the normal slope the score network must represent. It satisfies $\kappa \approx 1/s$ for $s^2 \gg \sigma^2$, peaks at $\kappa_{\max} = 1/(2\sigma)$ near $t \approx \sigma^2$, and falls back to 0 as $t \to 0$ (Figure 1a). In the singular limit the *peak* slope diverges, which is the precise sense in which infinite Fisher information bites the estimator.

**Samplers.** $S_{\text{trunc}}(t_0)$ integrates the learned reverse SDE from $T$ down to $t_0 > 0$ (Euler-Maruyama, $x \leftarrow x + (-\frac{1}{2}x + \hat{\varepsilon}/s)\, dt + \sqrt{|dt|}\, \xi$) and outputs $x_{t_0}$; all prior remedies, including the annealed schedules of Khelifa et al. [1], are of this class. $S_{\text{jump}}(\hat{t})$ integrates only to a moderate $\hat{t}$ and then outputs the Tweedie posterior-mean jump $\hat{x}_0 = (x_{\hat{t}} - s(\hat{t})\, \hat{\varepsilon})/a(\hat{t})$ [11, 14].

**The recursion.** For a distribution $p$ on $\mathbb{R}^d$ define the per-direction normal second moment $v(p) = \mathbb{E}_p[\operatorname{dist}(X,M)^2]/(d-k)$; true data has $v = \sigma^2$. Generation $g$ trains $\hat{\varepsilon}_g$ by denoising score matching on $\text{pool}_g = \lambda\,(\text{fresh real}) + (1-\lambda)\,(\text{generation } g{-}1)$ and samples; $v_g$ is the output value. Since $v$ is linear in the measure, $w_g := v(\text{pool}_g) = \lambda\sigma^2 + (1-\lambda)v_{g-1}$ exactly.

**Modeling assumptions (*we model*).** Our law rests on two assumptions about the trained pipeline that we *measure and cross-validate* but do not derive.

**Assumption 1** (scalar reduction)**.** *The per-generation map factors through the pool width:* $v_g = F_g(w_g)$. Why a scalar: *the collapse observable is a* variance *along the normal fibre, one number per normal axis by construction, so this is an order-parameter statement (the width closes on* *itself) rather than a claim that the network is scalar, in the same way an order parameter closes a mean-field recursion without the microscopic state being low-dimensional.* Validation: *fixed points measured in one experiment type predict those of an independent type within* 1 *to* 13% *across six* $(t_0, \lambda)$ *cells (Section 7).*

**Assumption 2** (concave response)**.** *$F_g(w) = a^2(t_0(g))\, w + s^2(t_0(g)) + e^2(w)$, where the sampling excess $e^2 : [0, \infty) \to (0, \infty)$ is nondecreasing and concave with $e^2(0) = e_0^2 > 0$.* In words: *the output width is the pool width shrunk by the forward map ($a^2 w$), plus the truncation remainder ($s^2$), plus a sampler-added excess $e^2$ that (i) never vanishes even for a perfectly thin pool ($e_0^2 > 0$, this is the floor), (ii) grows with a fatter pool but (iii) with diminishing returns (concave: doubling the pool width less than doubles the added excess).* Why concave: *the excess comes from the network failing to represent the steep part of the score, and a fatter pool is* easier *to represent (the required peak slope $\kappa_{\max} = 1/(2\sigma_{pool})$ falls), so each extra unit of pool width buys less extra error, exactly a saturating (concave) response.* Validation: *two independent estimators. The direct probe (training single-width pools and measuring $dv/dw - 1$; Appendix F) gives a local slope that is small and nonincreasing,* $+0.13 \to +0.05$*, across the upper half of the measured range, where all of the recursion's fixed points sit; its small magnitude is itself a measured near-cancellation of two $O(0.7)$ channels (Section 7). The recursion's own local convergence rates give an independent, indirect estimate that falls from* $+0.6$ *at the thinnest measured fixed-point pool ($\lambda = 0.75$) to $\approx 0$ at the fattest ($\lambda = 0.25$): the same saturating direction, though the two estimators disagree in magnitude where they overlap, and we treat the rate-based values as directional only (rate fits must separate a transient, a known failure mode of that estimator). At the probe's thinnest widths the slope estimates rise from* 0*, a regime the recursion never visits; the law's conclusions require only a unique downcrossing of $T(v) - v$, which the measured slopes ($\leq 0.13$ at every fixed point, against thresholds $\lambda/(1-\lambda) \geq 1/3$ at every plateau-forming fraction we test, $\lambda \geq 1/4$) guarantee.*

# 4 The law of collapse

Under Assumptions 1 and 2 the recursion is one-dimensional:

$$v_g = T(v_{g-1}) + s_g^2, \qquad T(v) = \lambda\sigma^2 + (1-\lambda)v + e^2\big(\lambda\sigma^2 + (1-\lambda)v\big), \tag{2}$$

with $s_g^2 := s^2(t_0(g))$ the truncation remainder. (The display takes $a^2(t_0) = 1$; the general case replaces $s_g^2$ by $\eta_g = s_g^2 + (a^2(t_0(g)) - 1)w_g$. Its second term $\to 0$ as $t_0 \to 0$ whenever $w_g$ stays bounded (part i), and at the small fixed $t_0$ we use it is $O(t_0 w_g)$, absorbed into the fitted law; in the divergent case (part ii) we work with the display form, where $\eta_g = s_g^2 \geq 0$ only strengthens the conclusion.) Let $c_\infty := \lim_{w\to\infty} e^{2\prime}(w)$ (exists by concavity) and define the *critical real-data fraction* $\lambda^* := c_\infty/(1 + c_\infty)$.

*How to read the theorem.* The per-generation update $T$ pushes the width toward a balance point. Part (i): if you keep enough fresh real data ($\lambda > \lambda^*$) there is a single stable plateau the width always settles to, from any start and any schedule. Part (ii): below that fresh-data threshold there is no plateau at all, the width runs off to infinity. (For our trained pipeline the measured saturated slope puts $\lambda^*$ at $\approx 0.03$–$0.05$; Section 7 locates it and exhibits both regimes.) Part (iii): if the sampler excess happens to be a straight line in the pool width, the plateau has a clean closed form. The one thing to notice is what $\lambda^*$ and $v_\infty$ depend on: the sampler excess $e^2$, *not* the truncation schedule, which is the whole point of Remark 1.

**Theorem 1** (law of the collapse; *we prove*)**.** *Fix $\lambda \in (0, 1]$.* ***(i)*** *If $\lambda > \lambda^*$, $T$ has a unique fixed point $v_\infty > 0$ solving $v = \lambda\sigma^2 + (1-\lambda)v + e^2(\lambda\sigma^2 + (1-\lambda)v)$, and for every $v_0 \geq 0$ and every truncation*

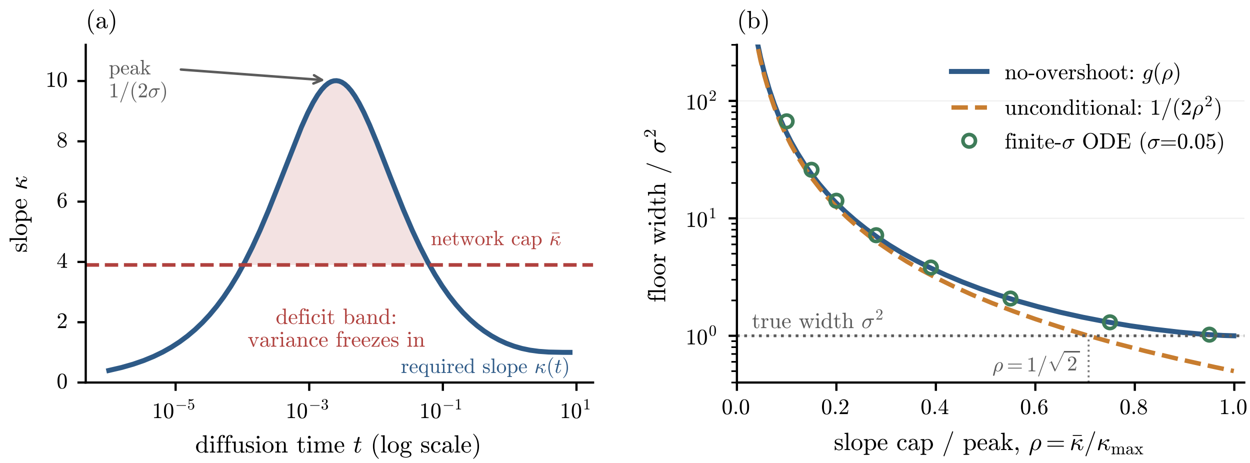


Figure 1: **The mechanism and the law.** (a) The required normal slope $\kappa(t) = s/\gamma^2$ is non-monotone: it peaks at $\kappa_{\max} = 1/(2\sigma)$ near $t \approx \sigma^2$ and falls back to zero. A network with ceiling $\bar{\kappa} < \kappa_{\max}$ under-contracts only on a *finite band* (shaded, the cap-to-curve deficit lens); variance frozen in the band cannot be removed below it (too few contraction e-folds remain). (b) The resulting floor in units of $\sigma^2$ as a function of $\rho = 2\sigma\bar{\kappa}$: closed-form no-overshoot law $g(\rho)$ (blue, Theorem 2), exact unconditional law $1/(2\rho^2)$ (orange). Green rings: direct finite-$\sigma$ ODE integration at $\sigma = 0.05$ swept across $\rho$ (self-checked against the appendix triples), sitting just above $g(\rho)$ and following the no-overshoot branch, the relative $O(\sigma^2/\rho^2)$ convergence the theorem predicts. The dotted vertical marks $\rho = 1/\sqrt{2}$, below which the floor exceeds the true width $\sigma^2$.

*schedule with $s_g^2 \to 0$ the iterates converge, $v_g \to v_\infty$, with local rate $\rho_{\text{loc}} = (1-\lambda)\big(1 + e^{2\prime}(w_\infty)\big) < 1$ (where $w_\infty = \lambda\sigma^2 + (1-\lambda)v_\infty$ is the fixed-point pool width; the strict inequality is forced by $T(0) > 0$, Appendix A).* ***(ii)*** *If $(1-\lambda)(1+c_\infty) > 1$, then $T(v) - v \ge T(0) > 0$ for all $v$ and $v_g \to \infty$: no plateau, for any schedule.* ***(iii)*** *In the affine special case $e^2(w) = e_0^2 + cw$: $\lambda^* = c/(1+c)$, $\rho_{\text{loc}} = (1+c)(1-\lambda)$, and $v_\infty = \big((1+c)\lambda\sigma^2 + e_0^2\big)/\big(\lambda(1+c) - c\big)$.*

*Proof.* Appendix A. Elementary but complete: concavity gives a unique downcrossing of $T(v) - v$; contraction above the fixed point plus a monotone barrier below give convergence under vanishing perturbations. □

*Remark* 1 (why annealing cannot help singular data). Truncation enters (2) only through the additive, schedule-killable $s_g^2$. The plateau $v_\infty$ is determined by $e^2$ alone, which no schedule touches. The remedy of Khelifa et al. [1] is the special case $e^2 \equiv 0$ (their Case 1 assumptions make the estimation term schedule-dominated), where indeed $v_\infty = \sigma^2$. Case 2 is precisely the regime where $e^2$ has a floor, the subject of the next section.

## 5 The capacity floor

We now prove that for manifold data the sampling excess $e^2$ of any truncated reverse-SDE sampler is floored by the normal slope its $\varepsilon$-predictor realises. No decomposition of the network is required. Let $X_t$ be the sampler's *own* ensemble at noise level $t$ and define the *realised normal slope*

$$\hat{\kappa}(t) := \frac{\operatorname{Cov}\big(X_t,\, \hat{\varepsilon}(X_t, t)\big)}{\operatorname{Var}(X_t)}, \tag{3}$$

the $L^2$ regression slope of the estimator on the normal coordinate under the law the sampler actually visits. For an affine estimator this is its slope; for an arbitrary nonlinear one it is the only functional of $\hat{\varepsilon}$ that the variance flow can see. The structural fact behind the floor (proof in Appendix B):

**Lemma 1** (variance ODE; *we prove*, exact)**.** *Under the reverse Euler scheme, for an* arbitrary *estimator $\hat{\varepsilon}$ and with $\hat{\kappa}$ as in (3), the ensemble normal variance obeys, in the small-step limit,*

$$-\frac{dV}{dt} = \Big(1 - \frac{2\hat{\kappa}(t)}{s(t)}\Big)V + 1. \tag{4}$$

*With the exact slope $\hat{\kappa} = \kappa$, $V(t) = \gamma^2(t)$ solves (4) exactly. The forcing $+1$ is the sampler's own injected noise. No linearisation is performed: to first order in the step a single Euler step moves the variance only through* $\mathrm{Cov}(X, \hat{\varepsilon}(X))$*, the $O(dt^2)$ remainder vanishing in the limit; equivalently, in continuous time Itô's formula yields (4) directly, with no expansion.*

*Remark* 2 (a checkable sufficient condition). The hypothesis of Theorem 2 constrains $\hat{\kappa}$, a regression slope. A pointwise bound implies it: if $\hat{\varepsilon}(\cdot, t)$ is $\bar{\kappa}$-Lipschitz in the normal coordinate then, for an i.i.d. copy $X'$ of $X_t$,

$$\mathrm{Cov}\big(X, \hat{\varepsilon}(X)\big) = \tfrac{1}{2}\mathbb{E}\big[(X - X')(\hat{\varepsilon}(X) - \hat{\varepsilon}(X'))\big] \le \tfrac{\bar{\kappa}}{2}\mathbb{E}\big[(X - X')^2\big] = \bar{\kappa}\mathrm{Var}(X),$$

so $\hat{\kappa}(t) \le \bar{\kappa}$ under *any* law. Lipschitz control is therefore sufficient but not necessary; we state the theorem on $\hat{\kappa}$ because it is the weaker hypothesis and because it is what (4) actually contains.

This is the precise sense in which the floor is *not* an artifact of linearizing the score. An earlier version of this argument split $\hat{\varepsilon}$ into an affine $L^2(p_t)$ fit plus an orthogonal residual and invoked the normal equations to discard the residual to first order. That step is no longer needed, and with it goes its one soft spot: the normal equations hold under $p_t$, whereas the sampler's own ensemble deviates from $p_t$. Equation (4) is exact for the fully nonlinear network precisely because $\hat{\kappa}$ is defined under the sampler's law from the outset. The crux is the geometry of the required slope:

**Lemma 2** (finite band; *we prove*)**.** *$\kappa(t)$ rises from 0, peaks at $\kappa_{\max} = 1/(2\sigma)$ at $t \approx \sigma^2$, and falls back to 0 as $t \to 0$. Hence for any ceiling $\bar{\kappa} < \kappa_{\max}$ the deficit set $\{t : \kappa(t) > \bar{\kappa}\}$ is a finite band; below it the exact slope is representable again, but only $2\log(1 + \rho^2/4) \le \rho^2/2$ contraction e-folds remain ($\rho := 2\sigma\bar{\kappa}$), so variance frozen in the band persists to $t_0 \to 0$.*

In the singular limit the whole flow collapses onto a one-parameter family (Appendix C for proofs and error terms):

**Lemma 3** (singular-limit ODE; *we prove*)**.** *Rescale $t = \sigma^2 u$, $V = \sigma^2 W$, and write the slope envelope as $\hat{\kappa}(t) = (1/\sigma)m(u)$ with $m(u) = \rho/2$ (unconditional envelope) or $m(u) = \min\{\sqrt{u}/(1+u), \rho/2\}$ (no-overshoot envelope). Then (4) becomes, as $\sigma \to 0$ at fixed $\rho$,*

$$-\frac{dW}{du} = 1 - \frac{2m(u)}{\sqrt{u}}\,W, \tag{5}$$

*a $\sigma$-free equation; the uncapped equation has exact solution $W = 1 + u$ (the rescaled true marginal), and deviations contract quadratically under downward integration. Consequently $\lim_{t_0 \to 0} V(t_0)/\sigma^2 = g_{\mathrm{class}}(\rho)\big(1 + O(\sigma^2/\rho^2)\big)$ (a* relative *correction: the boundary-layer perturbation scales with the leading term $W \sim \rho^{-2}$, so the additive gap to $g$ is $O(\sigma^2/\rho^4)$, as the finite-$\sigma$ values confirm):* ***universality in $\rho$ is a theorem****.*

*How to read the theorem.* There is one dimensionless knob, $\rho = \bar{\kappa}/\kappa_{\max} = 2\sigma\bar{\kappa}$: how close the network's slope ceiling gets to the peak slope the data demands. The floor is a universal function of $\rho$ alone. Version (U) makes no assumption on the network and gives the clean bound $1/(8\bar{\kappa}^2)$; version (N) additionally assumes the network never over-shoots the true slope (which is what we measure) and gives a slightly tighter constant. The two constants differ by only 5.5%, so for intuition either one is "the" floor. The single takeaway: when $\rho < 1/\sqrt{2}$ the floor sits strictly above the true width $\sigma^2$, i.e. the sampler is stuck too fat, and it gets relatively worse the thinner the manifold.

**Theorem 2** (closed-form capacity floor; *we prove*)**.** *Let the trained score's normal slope satisfy $\hat{\kappa}(t) \le \bar{\kappa} < \infty$ (a finite normal Lipschitz constant: a bound on slope, not on parameter count), and set $\rho = 2\sigma\bar{\kappa}$. Then for every truncation time $t_0 > 0$ and every schedule:*

***(U) Unconditional.*** *With no further assumption (arbitrary overshoot permitted),*

$$V(t_0) \;\ge\; \sigma^2 \frac{1}{2\rho^2}\,(1+o(1)) \;=\; \frac{1}{8\bar{\kappa}^2}\,(1+o(1)) \qquad (\sigma \to 0), \tag{6}$$

*where $1/(2\rho^2)$ is the* exact *value of* (5) *with $m \equiv \rho/2$: $W(0^+) = \int_0^\infty e^{-2\rho\sqrt{u}}\,du = 1/(2\rho^2)$. The floor exceeds $\sigma^2$ whenever $\rho < 1/\sqrt{2}$, i.e. $\bar{\kappa} < \kappa_{\max}/\sqrt{2}$.*

***(N) No-overshoot.*** *If additionally $\hat{\kappa}(t) \le \kappa(t)$ (the empirically observed profile; the network never over-contracts), the tighter floor $V(t_0) \ge \sigma^2 g(\rho)(1+o(1))$ holds with the explicit closed form: writing $q = \sqrt{1-\rho^2}$, $x_\mp = (1 \mp q)/\rho$, $u_\mp = x_\mp^2$,*

$$W_{\text{exit}} = (1+u_+)e^{-4q} + \frac{(3-2q)-(3+2q)e^{-4q}}{2\rho^2}, \qquad g(\rho) = 1 + \frac{W_{\text{exit}} - (1+u_-)}{(1+u_-)^2}. \tag{7}$$

*$g(\rho) > 1$ for every $\rho \in (0,1)$ with $g \to 1$ only as $\rho \to 1^-$, and $g(\rho) = C_0/\rho^2 + O(1)$ as $\rho \to 0$ with $C_0 = (1+3e^{-4})/2 = 0.5275$, i.e. the deterministic single-pass floor $\Phi_{\det} := \lim_{t_0\to 0} V(t_0) = \sigma^2 g(\rho)$ satisfies $\Phi_{\det} \to \big((1+3e^{-4})/8\big)\bar{\kappa}^{-2} = 0.1319\,\bar{\kappa}^{-2}$.*

*The two asymptotic constants, $1/8$ and $(1+3e^{-4})/8$, differ by 5.5%: the class distinction is quantitatively minor. The closed forms match direct integration of* (5) *to $\le 2.4 \times 10^{-5}$ across $\rho \in [0.1, 1]$, and the finite-$\sigma$ ODE converges to them from above at the stated relative $O(\sigma^2/\rho^2)$ rate.*

**Corollary 1** ($\sigma \to 0$: the answer to Case 2; *we prove*)**.** *Hold $\bar{\kappa}$ fixed and let $\sigma \to 0$ ($J \to \infty$). Then unconditionally $V/\sigma^2 \ge (1+o(1))/(8\bar{\kappa}^2\sigma^2) \to \infty$: a truncated sampler held at a fixed slope ceiling collapses* relatively *without bound, for every annealing schedule. Conversely, bounded relative collapse $V/\sigma^2 \le C$ requires*

$$\bar{\kappa} \;\ge\; \frac{1}{2\sigma\sqrt{2C}} \;=\; \Omega(1/\sigma) \;=\; \Omega(\sqrt{J}). \tag{8}$$

*Truncated sampling of singular data demands a slope budget growing with the singularity. This constrains the normal Lipschitz norm: a fixed architecture can meet it only by enlarging its weights, and whether training in fact does so is the empirical question the $\bar{\kappa}(w)$ probe addresses. We thank Wei Huang for pressing this distinction.*

**What each ingredient does (*we observe*).** The ablation of Figure 2 separates the roles exactly as (4) dictates: the sampler's own noise (the $+1$) is load-bearing for the floor's *existence*: switching it off sends $V \to 0$ (point collapse, the deterministic horn), while the finite ceiling $\bar{\kappa}$ makes the floor *larger than* $\sigma^2$ (the stochastic horn). The two branches bracket the target from both sides; no choice of the sampler noise magnitude attains it.

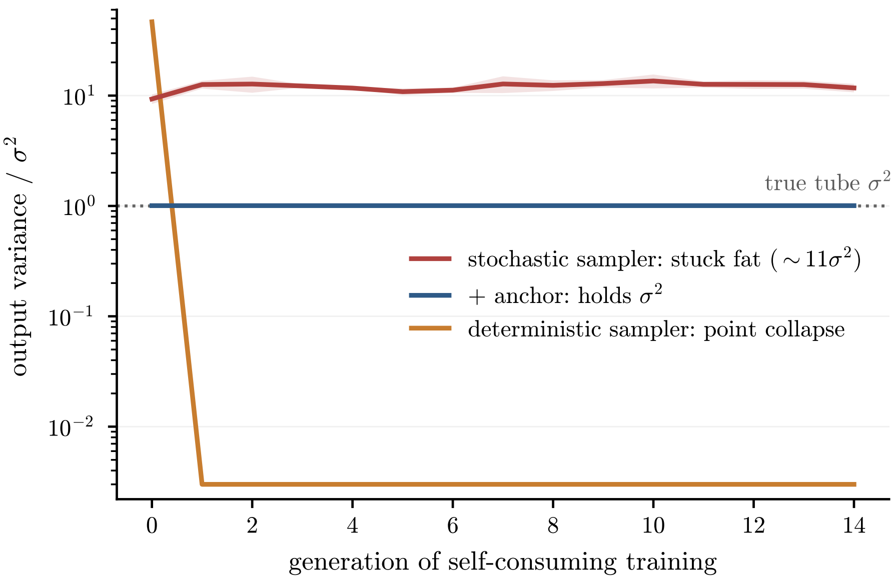


Figure 2: **The two-horned trap (*we observe*).** Self-consuming recursions at fixed $t_0 = 10^{-4}$, no anchor, 3 seeds. With the reverse-SDE noise on, the recursion sticks fat at 10.8 to 13.2 $\sigma^2$ (red). With the noise zeroed (deterministic sampler), it collapses to a *single point* within one generation (orange; verified loss of all 2-D spread, no numerical overflow). Neither branch reaches the tube ($\sigma^2$, dotted); the anchored recursion (blue, Section 6) holds it.

**Confronting the floor with a real network (*we observe*).** For our standard pipeline the measured slope profile tracks $\kappa(t)$ faithfully at benign times ($\hat{\kappa}/\kappa = 0.95$ to 0.97 for $t \geq 0.1$) and then saturates while $\kappa$ continues to $\sim 10$. *Which* ceiling one reads off depends on the law the regression is taken under, and (3) settles that: the hypothesis is the slope under the sampler's *own* ensemble. Probing the forward process instead returns a systematically larger ceiling. Measured on the same networks across five tube widths (`kbar_sampler_law.py`; $\sigma = 0.03$ to 0.13), the sampler-law ceiling is 0.83 to 0.87 of the forward-process one, monotone in $\sigma$ and without exception.

*The gap is curvature, not saturation (*we observe*).* The natural explanation — that the sampler's ensemble is fatter than $p_t$ and so probes the response into its flattening tails — is wrong, and the geometries of §7 refute it. Re-measuring all of them under both probes (`capacity_domains_samplerlaw.py`) gives a probe ratio of 0.995–0.998 on the straight segment and 0.975–0.988 on the codimension-9 ring, whose slope is read on its eight *out-of-plane* axes: both flat normal coordinates, both agreeing to within 1%. The sampler ensemble is fatter in those settings too, so fattening alone cannot be the mechanism. Where the normal coordinate is *curved* the probes separate: 0.830–0.841 on the ring tube, whose radial normal curves in one direction, and 0.740–0.746 on $S^2$, which curves in two. The ordering is consistent across all five widths, though the sphere's deficit is 1.5–1.8× the ring's rather than the 2× strict proportionality in the number of curved directions would give, so we report the ordering and not a law. Two consequences. The flat tube the theorem idealises is exactly the case where the two readings coincide, so (3) and the forward-process probe agree in the setting Theorem 2 assumes; and the discrepancy on ring data is a curvature artifact of the probe, of the same origin as the $O(s^2/\tau)$ term §8 declines to carry. Reassuringly, the relative degradation is untouched: the drop across $\sqrt{w}$: $0.03 \to 0.13$ reads 42.3% against 42.2% on the segment, 29.8 against 29.2 on the sphere, and 48.7 against 48.0 on the codimension-9 ring, so every trend claim in §7 stands under either probe. Since the floor scales as $\bar{\kappa}^{-2}$, the quantity the theorem constrains gives the *larger* bound. Over six seeds at $\sigma = 0.05$: $\bar{\kappa} = 3.26$ rather than 3.90, so the envelope $\Phi_{\mathrm{det}}$ reads $5.49\,\sigma^2$ rather than $3.83\,\sigma^2$ (Figure 3), accounting for 81% of the measured single-pass floor against 58% read under $p_t$, with no seed falsified. The correct reading is also the more reproducible one: its seed spread is 77–85% against 42–72% for the forward-process probe, which is what one expects once the probe and the dynamics are evaluated under the same law. The exact-score control reproduces $\gamma^2(t_0)$ to all displayed digits.

*Where the comparison stops working (*we observe*).* The margin narrows monotonically as the tube fattens: 77–85% of the measured floor at $\sigma = 0.05$, 90–97% at $\sigma = 0.10$, and 97–103% at $\sigma = 0.13$, where the envelope exceeds the measured floor in three of six seeds. We report this rather than trim the range. It is not evidence against Theorem 2, whose chain we verify symbolically (Appendix C); it is the flat-tube idealisation expiring where Section 8 says it will, curvature entering at $O(s^2/\tau)$ and not carried in the constants. At $\sigma = 0.13$ on a radius-2.5 ring that correction is no longer negligible. The monotone narrowing is the signature of an asymptotic bound pushed past its regime, not of a defect in the derivation, and it places the honest validity edge of the *numerical* comparison between $\sigma = 0.10$ and $\sigma = 0.13$. We checked the obvious alternative first: taking $\bar{\kappa}$ as the maximum over the whole deficit band rather than over five probe times moves it by $<0.02$, and leaves two of six seeds violating.

**What is left over (*we observe*).** Re-running the sampler's own discrete variance recursion with measured inputs now resolves the remainder into *one* channel rather than two. Earlier versions of this measurement carried an **ensemble-displacement** term, worth $+2.6\,\sigma^2$, for precisely the probe mismatch just described; under (3) that is not a correction to the bound but part of the hypothesis, and it is no longer counted separately. What remains is **profile shape**: the measured $\hat{\kappa}(t)$ sags $\approx 0.5$–$0.7$ below its own ceiling inside the deficit band, worth $+0.7\,\sigma^2$ at the data width. Beyond it the affine recursion driven by the measured ensemble slope reproduces the sampler to $-0.4\,\sigma^2$: the nonlinear remainder is small. **Interventional test.** The law survives moving the ceiling: five training protocols ($t$-importance-sampling into the band, $4\times$ steps, a higher-frequency time embedding, and combinations) move $\bar{\kappa}$ from 3.5 to 7.5; the measured floor drops from 7.4–7.8 $\sigma^2$ at the two low-$\bar{\kappa}$ protocols into the 2.5–4.9 $\sigma^2$ range at the three high-$\bar{\kappa}$ ones (a group-level fall with protocol-dependent scatter, not a clean $1/\bar{\kappa}^2$ trace), and — the load-bearing statement — every one of the five sits strictly above its own proven bound.

That statement is the one most exposed by (3), since a lower sampler-law ceiling *raises* each arm's bound, so we re-measured all five under both probes (`ceiling_origin_samplerlaw.py`, three seeds per arm). It survives: no arm is violated, and the worst now sits at 78% of its measured floor against 60% read under $p_t$. The probe ratio is not a constant of the geometry but a property of the trained network, running 0.754 to 0.853 across the five and splitting cleanly by training protocol — 0.843–0.853 for the three arms drawing noise levels uniformly, 0.754–0.769 for the two importance-sampling half their draws into the deficit band, which trains the network harder exactly where the sampler's ensemble is fattest relative to $p_t$. A single scale factor is therefore not a substitute for the measurement.

**What sets the ceiling (*we observe*).** The interventions also identify the ceiling's origin. Raising the time-embedding frequency alone does nothing ($\bar{\kappa} = 3.5$): expressivity in $t$ is not binding. Importance-sampling training times into the band ($\bar{\kappa} \to 6.7$) or simply training $4\times$ longer ($\bar{\kappa} \to 6.4$) moves it strongly: under the uniform $t$-draw the deficit band holds $<1\%$ of the training measure (and $t < 10^{-3}$ is never drawn at all, though the sampler visits it), so the ceiling is *gradient starvation* — an optimization constant, not an architectural one, consistent with its independence of sample size

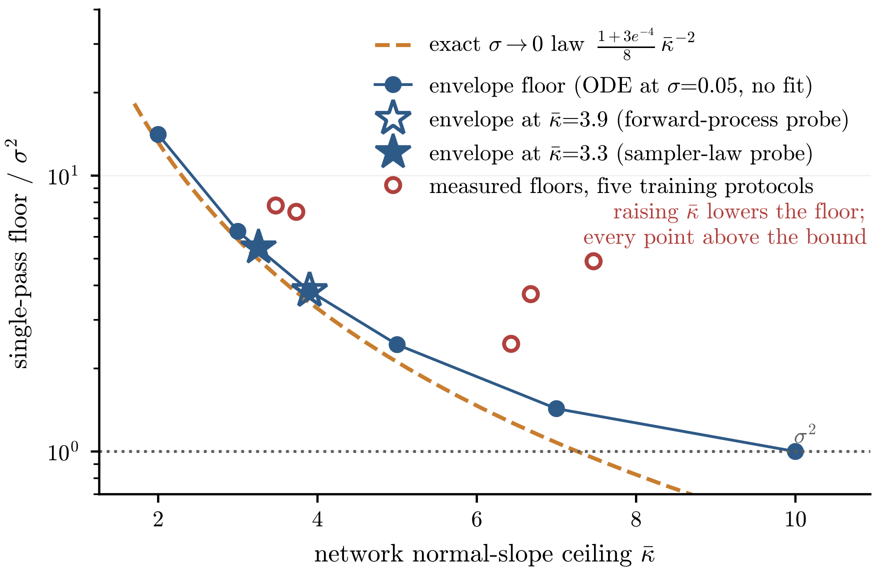


Figure 3: **The capacity floor: predicted, measured, and moved.** Blue line with markers: finite-$\sigma$ variance-ODE envelope floor at $\sigma = 0.05$ (no fit). Dashed: the exact $\sigma \to 0$ law $\big((1+3e^{-4})/8\big)\bar{\kappa}^{-2}$ of Theorem 2. Stars: the envelope bound at the measured ceiling, read two ways. Hollow, under the forward process ($\bar{\kappa} = 3.9$, bound $3.8\,\sigma^2$); filled, under the sampler's own ensemble (3), which is what the theorem constrains ($\bar{\kappa} = 3.3$, bound $5.5\,\sigma^2$; six seeds). Both are proven lower bounds; the second is the tighter, accounting for 81% of the directly measured single-pass floor against 58%. The measured anatomy of the remainder (in-band profile sag) is in the text, as is the tube width at which the comparison expires. Open circles: the interventional test — five training protocols move the measured ceiling from 3.5 to 7.5; the measured floor falls from $\sim$7.6 $\sigma^2$ (low $\bar{\kappa}$) into the 2.5–4.9 $\sigma^2$ range (high $\bar{\kappa}$), with protocol scatter, and every point stays strictly above its bound.

and its degradation on fatter pools (a fatter tube spreads the same starved gradient over an easier, wider band). The Case-2 conclusion is unchanged — as $\sigma \to 0$ the demanded peak $1/(2\sigma)$ diverges, so any fixed training budget caps $\bar{\kappa}$ — but the constant is steerable: band-weighted noise-level sampling buys a 2–3× lower per-pass floor at fixed size. It cannot reach $\sigma^2$, and the recursion compounds whatever remains, so anchoring is still the fix.

**An honest gap, closed in three measured steps (*we model*).** Iterating the floor through the recursion (every generation's pool is at least as wide as $\lambda\sigma^2 + (1-\lambda)\Phi_{\text{det}}$) with the network's *clean-tube* ceiling $\bar{\kappa} = 3.9$ gives a deterministic fixed point $v^*_{\text{det}} = 4.5\,\sigma^2$. The measured recursion floor is $\approx 11\,\sigma^2$, a factor 2.4 above; our theorem claims only a *lower bound* on the recursion (respected). We account for that factor in three measured steps, each a directly logged measurement.

*Step 1, capacity degradation.* The single-width probe (Appendix F) shows the ceiling itself degrades on fatter training pools, from 3.7–4.3 at $\sqrt{w} = 0.03$ down to 2.8–3.0 at $\sqrt{w} = 0.13$ (ranges over three training seeds), and the degradation replicates under a half-width network, under a different ring radius (slopes $-9.4$ to $-13.0$ across all five measurements), across three further synthetic geometries — a flat segment (the flat-tube idealization made exact, so the degradation does not require curvature; 42% drop over the same width range), a sphere in $\mathbb{R}^3$ (30%), and a circle in $\mathbb{R}^{10}$ (codimension 9; 49%) — and, crucially, on *genuine MNIST* $8 \times 8$ *pixel data* (whose digit manifold has a measured intrinsic normal width $\sigma_0 \approx 0.072$, thin against its tangent scale $\approx 0.41$):

with the identical extraction protocol and per-point $k$NN-PCA normal frames, $\bar{\kappa}$ falls $2.0 \to 1.6$ as $\sqrt{w}$ runs $0.071 \to 0.231$ (21%), tightly fit by $\bar{\kappa}(w) \approx 2.2 - 2.7\sqrt{w}$ (max residual 0.017, two seeds). The degradation is therefore not a synthetic-tube artifact. Constants are geometry-specific. Over this range the points are well described by the two-parameter fit $\bar{\kappa}(w) \approx 4.4 - 11.6\sqrt{w}$, an *empirical fit, not a derived form*; nothing below depends on the choice, since refitting linear, logarithmic, or quadratic, or replacing the fit with a fit-free monotone interpolation of the raw points, moves the resulting fixed point by at most 10% (the computation uses the fit only as an interpolant inside the measured range).

*Under the law the theorem constrains.* The probe above reads the ceiling off the forward process; (3) asks for it under the sampler's own ensemble. Re-running the identical protocol with both probes on the same networks (`poolwidth_probe_samplerlaw.py`; five widths, three seeds each) reproduces every published forward-process ceiling to three decimals, and gives a sampler-law ceiling 0.830 to 0.841 of them. That band is narrow but not monotone in $\sigma$, unlike the $t_0 = 10^{-4}$ probe of Section 5, whose ratio does rise monotonically; this probe inherits its parent's truncation $t_0 = 5 \times 10^{-3}$, and the two agree on the level of the ratio while differing on its weak trend, so we read a constant $\approx$0.83 here and do not claim a direction. The degradation law is not merely rescaled but *flatter*, $\bar{\kappa}(w) \approx 3.80 - 10.25\sqrt{w}$ against $4.54 - 12.17\sqrt{w}$ (max residual 0.031 and 0.047), so no single factor converts one into the other. Propagated through the same closure, the deterministic fixed point moves from $5.8\,\sigma^2$ to $10.0\,\sigma^2$.

*Step 2, the self-consistent closure.* Feeding the degradation back, $v \leftarrow \Phi_{\text{det}}\big(w(v), \bar{\kappa}(w(v))\big)$, moves the fixed point from 4.5 to 5.6–7.1 $\sigma^2$ at $t_0 = 10^{-4}$ (the range across the three probe seeds' fits), with the resulting pool width $\sqrt{w^*} \approx 0.091$–$0.101$ *inside* the measured support (interpolation, not extrapolation). The closure assumes the degradation depends on the pool only through its width $w$, while the recursion's actual pools are two-component mixtures (clean tube plus fattened returns); testing this directly by training on the fixed point's own mixture composition gives $\bar{\kappa}^{\text{mix}} = 3.37 \pm 0.15$ against the single-tube prediction 3.34, so the width-only reduction survives its first direct test. This closure shrinks the unexplained factor from 2.4 to 1.5–2.0 — a quarter to a half of the logarithmic gap.

*Step 3, the measured remainder.* For what remains we first say what it is *not*: iterating the map with the measured generation-to-generation fluctuations gives a Jensen bias of only $0.006\,\sigma^2$, and applying the measured $t \geq 0.1$ slope-tracking ratio (0.95–0.97) to the whole profile moves the fixed point by only $+0.3$–$0.4\,\sigma^2$. The remainder is then *measured*: the profile-sag and ensemble-displacement channels of the floor anatomy above, re-measured across six pool widths spanning the fixed point, lift the per-pass map to its directly measured values, and iterating that measured map gives a fixed point of $9.9\,\sigma^2$ against the measured $10.8$–$13.2\,\sigma^2$ — the once factor-2.4 discrepancy is now 1.1–1.3, at the level of protocol differences (mixture pools, per-generation sample counts, seed scatter). What stays underived are the constants themselves: $\bar{\kappa}$, its degradation law, and the saturation profile.

*Two steps, not three.* Of the two channels Step 3 applies, ensemble displacement is the probe mismatch of Section 5: under (3) it is not a correction to the bound but part of the hypothesis. Measuring $\bar{\kappa}$ under the sampler's own law therefore reaches $10.0\,\sigma^2$ in Step 1 alone, against the $9.9\,\sigma^2$ Step 3 reaches by applying it afterwards — the same destination by a shorter route, and to 1% rather than to the level of protocol differences we had expected to have to invoke. The residual factor of 1.1–1.3 against the measured $10.8$–$13.2\,\sigma^2$ is unchanged in size and character; what changes is that one of the two channels no longer has to be carried.

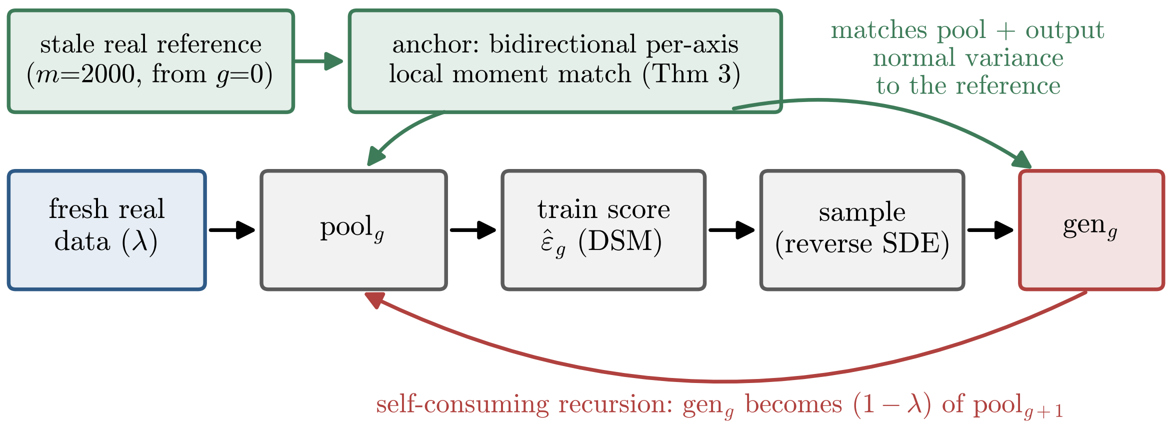


Figure 4: **Where the anchor sits.** Each generation, the pool (and, for the jump pipeline, the sampler output) is passed through a bidirectional per-axis local moment match against a *stale* reference of $m = 2{,}000$ real samples drawn once at $g = 0$. No provenance labels are needed; the reference is never refreshed.

## 6 The anchoring fix

Corollary 1 says the singular regime cannot be won by scaling the model or tuning the schedule. It can be won by *anchoring the recursion to data*.

**The operator.** For each point, find its $k$ nearest neighbors in the reference; take the local PCA frame (top-$k_{\text{dim}}$ directions tangent, remainder normal); rescale the point's normal coordinates so that the pool's local normal second moments equal the *reference's* local normal second moments in the *same estimated frames*. The match is *bidirectional* (it contracts fat directions and inflates thin ones) and *per-axis*. Both properties are forced by the theory: the sampler's error is signed (fat from injection, thin from posterior-mean sharpening; Proposition 1 in Appendix D) and anisotropic whenever the normal spectrum straddles $s^2(\hat{t})$, so any scalar or one-sided matcher mis-corrects some directions (*we observe*: a scalar matcher leaves MNIST-latent thin ratios at 0.62; the per-axis matcher achieves $|\text{thin} - 1| \leq 0.012$).

**Lemma 4** (shared-frame cancellation; *we prove* to first order)**.** *Let both reference and pool normal energies be measured in the same kNN-PCA frames estimated from the reference. Then the matched pool satisfies, per normal direction,* $v(\text{matched}) = \hat{\sigma}^2_{\text{ref}} + O_p(m^{-1/2}) + \Delta_{\text{leak}}$*, where the frame-error leakage* $\Delta_{\text{leak}}$ *afflicts pool and reference* identically *at first order in the frame perturbation and cancels; standard local-PCA rates [15, 16] enter only at second order. Every quantity in the bound is intrinsic* $(k, \tau, \sigma, m)$*; the ambient dimension* $d$ *appears nowhere.*

*How to read the theorem.* The collapse of Theorem 1 happens because generation $g$ inherits generation $g-1$'s width: the map $F_g$ has a $v_{g-1}$ inside it, so errors compound. The matcher overwrites the pool's normal width with a fixed number (the reference's) *before* training, so the input to $F_g$ no longer depends on $v_{g-1}$. The recursion becomes a constant sequence, there is nothing left to compound, and the critical-fraction threshold of Theorem 1 disappears entirely. The only error left is the one-time cost of estimating the reference width from a finite sample, and that does not grow with generations.

**Theorem 3** (anchoring makes the recursion memoryless; *we prove* given Lemma 4 and Assumption 1)**.** *Insert the matcher after pooling (and after sampling, for the jump pipeline). Write* $\varepsilon_m := O_p(m^{-1/2}) + \Delta_{\text{leak}}$ *for the matcher error of Lemma 4, a quantity that does not grow with* $g$*. Then the training input has normal variance* $\hat{\sigma}^2_{\text{ref}} + \varepsilon_m$ regardless of $v_{g-1}$*: the map* (2) *loses its* $v$*-argument,*

$$v_g = F_g\big(\hat{\sigma}^2_{\text{ref}} + \varepsilon_m\big),$$

*a constant sequence up to the non-compounding* $\varepsilon_m$*. Consequently: no fixed-point equation,* ***no critical fraction, and stability for every*** $\lambda > 0$*; and with* $S_{\text{trunc}}$ *the dynamics reduce to truncation-only, restoring the* conclusion *of Khelifa et al. [1]'s Theorem 4.1 in the singular regime (the data remains singular; what anchoring removes is the compounding term that their hypotheses excluded).*

**Why this is not a replay buffer.** Replay, data-accumulation, and data-refresh remedies [4, 8] fight collapse by *re-inserting real examples* into the pool: they preserve *samples*. The anchor preserves something weaker and more portable, the *local normal moments* (per-axis widths) of a small stale reference, and imposes them on *whatever the model currently generates*. It never adds a reference point to the training pool, uses no provenance labels, and reuses the same $m$=2,000 points for every generation. The distinction is not cosmetic. A replay buffer of size $m$ still lets the $(1-\lambda)$ synthetic majority set the pool's fine-scale statistics, so the floor of Section 5 still governs that majority; anchoring instead *overwrites* the fine-scale statistics of the majority itself, which is why it removes the floor rather than diluting it. In the language of the law (Theorem 1): replay changes $\lambda$; anchoring changes the response map $F_g$. We test this at the anchor's exact information budget (*we observe*): inserting the anchor's own stale reference ($m = 2{,}000$ real points, drawn once at $g = 0$, never refreshed) *into the pool* every generation of the head-to-head protocol — raising the effective real fraction to $\lambda_{\text{eff}} = 2/3$, with synthetic data now a 1/3 *minority* — moves the 20-generation plateau only from $9.8 \pm 1.7$ to $8.5 \pm 1.1\,\sigma^2$ (5 seeds, same annealed schedule), the small dilution the law predicts for a larger $\lambda$; the anchor, using the same 2,000 points as a moment reference instead, holds $1.01 \pm 0.02\,\sigma^2$. Replay dilutes the floor; anchoring removes it.

**Attribution, stated plainly.** In our pipeline the *matcher* is the floor-breaker; the Tweedie jump is a *raw-error reducer*. The jump's raw output is itself fat (7 to $20 \times \sigma^2$ across settings); its value is a $\approx 2\times$ smaller one-shot raw error and $\approx 3\times$ smaller unanchored plateau (21 vs $68\,\sigma^2$ in raw normal variance, at $\lambda = \frac{1}{2}$ and matched moderate truncation $t_0 = \hat{t} = 0.05$; the standard plateau includes the truncation remainder $s^2(0.05)/\lambda \approx 39\,\sigma^2$ that the jump removes, and the jump plateau is itself predicted by Theorem 1 as $e^2_{\text{jump}}/\lambda$), plus smaller post-match tails. Post-anchor, the standard and jump samplers converge to the same variance; we report both.

**Dimension scaling (*we observe*).** Lemma 4's $d$-independence is a postdiction: across ambient $d = 8 \to 128$ (curved 2-manifolds, fixed tube), the *raw* one-shot floors grow steeply ($\sim d^{0.76}$ standard, $\sim d^{0.95}$ jump, the collapse itself), while the *matched* floors are flat ($\sim d^{0.1}$), and anchored recursions are flat for 8 generations at both $d = 8$ and $d = 128$.

# 7 Experiments

All core experiments (ring, curved manifolds, MNIST, law identification) are CPU-only ($\leq 12$ threads) and use small MLPs (3 hidden layers, width 256, SiLU, sinusoidal time embedding) with denoising score matching and geometric reverse-time grids; the CIFAR-10 scale-up alone uses a

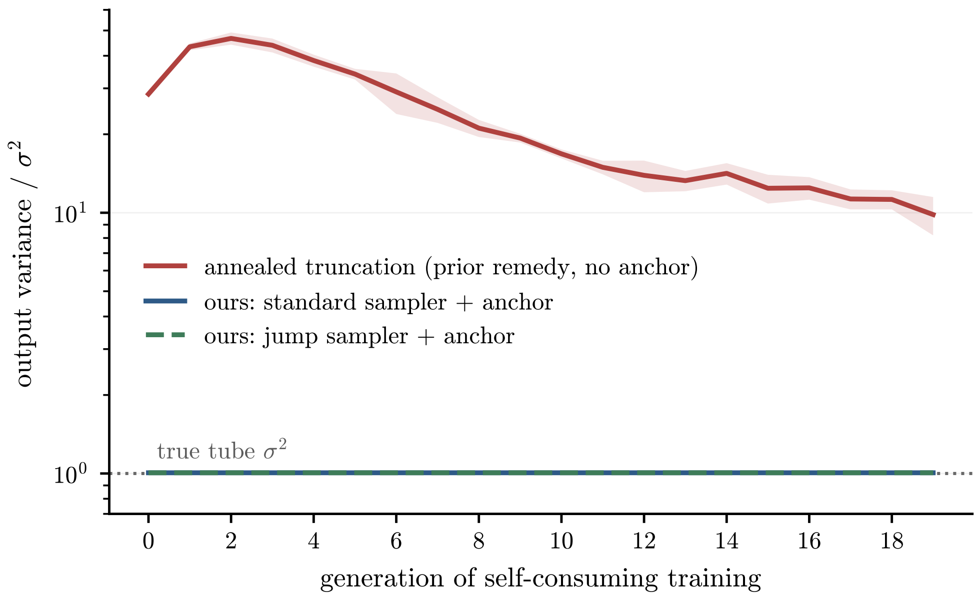


Figure 5: **Head-to-head vs. the state-of-the-art remedy (*we observe*).** Ring manifold, $\sigma = 0.05$, $\lambda = \frac{1}{2}$, 20 generations, 5 seeds (mean $\pm$ s.d.). Red: annealed truncation $t_0 : 0.05 \to 10^{-4}$, no anchor [1]; it rises to $47\sigma^2$, then thins only to $9.8\sigma^2 \pm 1.7$. Blue/green: the same recursion with the anchor (standard and jump samplers): $1.01\sigma^2 \pm 0.02$, flat for 20 generations.

$\sim$7M-parameter convolutional denoiser on a single consumer GPU (mixed precision). Every long run streams to an append-only log and is exactly resumable. Full protocols, seeds, and a script-to-claim map are in the released code (Section 9).

**Protocols.** *Ring*: radius 2.5, $\sigma = 0.05$ ($\sigma^2 = 0.0025$), 4,000 samples/generation, 2,000 training steps. *Head-to-head* (Figure 5): $G = 20$, $\lambda = \frac{1}{2}$, 5 seeds; arm A anneals $t_0 : 0.05 \to 10^{-4}$ geometrically (the remedy of 1); arms B/C add the anchor to the standard/jump samplers. *Two-horned ablation* (Figure 2): fixed $t_0 = 10^{-4}$, noise on/zeroed, $G = 15$, 3 seeds. *Replay control*: head-to-head protocol with the anchor's $m$=2,000 stale reference inserted into the pool each generation instead of used as a moment reference ($G = 20$, 5 seeds). *Critical fraction*: fixed $t_0 = 10^{-4}$, $\lambda \in \{0.02, 0.25\}$, 3 seeds each. *Pixel MNIST* (Figure 6): $8 \times 8$ pixels, $G = 8$, 3 seeds, local matcher with $k = 96$, $k_{\text{dim}} = 12$. *Law identification*: fixed-$t_0$ recursions at $t_0 \in \{0.02, 0.05\}$, $\lambda \in \{0.25, 0.5, 0.75\}$; the affine law fitted at one $t_0$ predicts fixed points at the other within 1 to 13% (Assumption 1 validation); the direct probe slope of $e^2$ is small and saturating ($+0.13 \to +0.05$ over the upper measured range), the net of a measured deficit channel $-0.68$ and injection channel $+0.73$, and the recursion's own convergence rates independently give a slope falling from $+0.6$ (thin fixed-point pools, $\lambda = 0.75$) to $\approx 0$ (fat, $\lambda = 0.25$) (Assumption 2 validation); a refinement using the probe's own capacity measurements splits the injection channel further (Appendix F).

**The critical fraction, located and exhibited (*we observe*).** The probe's saturated slope, $c_\infty \approx 0.03$–$0.05$, puts the critical fraction at $\lambda^* = c_\infty/(1 + c_\infty) \approx 0.03$–$0.05$ for this pipeline — small, so the collapse is self-limiting at any practically relevant fresh-data fraction, and the theorem's content is that $\lambda^* > 0$ at all. A recursion at $\lambda = 0.02$ (at or below the estimate; fixed $t_0 = 10^{-4}$, 3 seeds, $G = 25$) exhibits the subcritical regime directly: the width climbs monotonically through $69 \pm 16\,\sigma^2$ at $g = 24$, roughly $10\times$ its early value, with no sign of flattening — while a matched $\lambda = 0.25$ control plateaus at $16.5 \pm 2.2\,\sigma^2$ by generation $\sim$5. (A finite horizon cannot distinguish

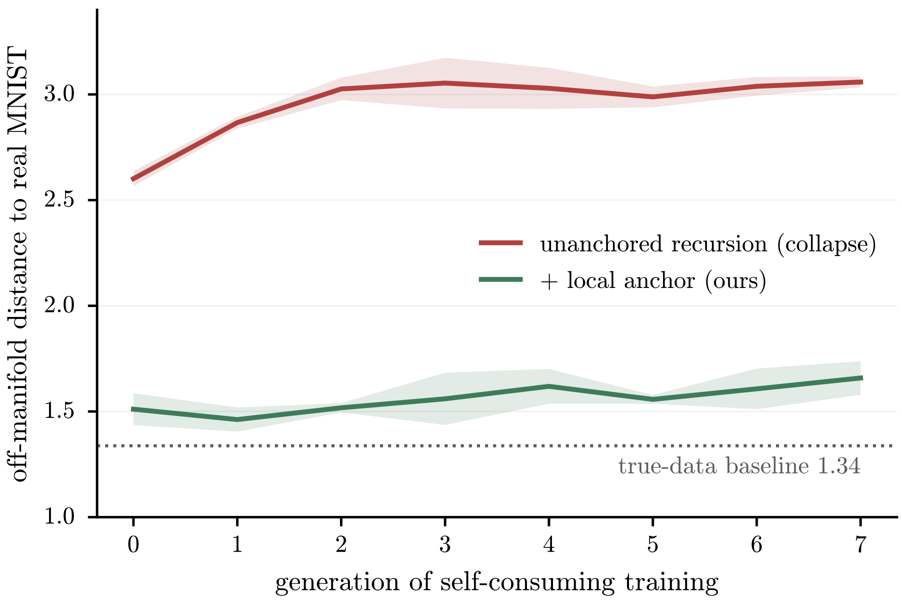


Figure 6: **Real pixels (*we observe*).** Self-consuming recursion on $8 \times 8$ MNIST (64-dimensional pixel space), $\lambda = \frac{1}{2}$, 8 generations, 3 seeds. Off-manifold distance = mean nearest-neighbor distance to a held-out real test set. Unanchored drifts to 3.06 (2.3× the true-data baseline 1.34); anchored holds 1.66 (1.24×), closing ∼81% of the gap. The mild upward creep of the anchored arm reflects the local-chart approximation on a genuinely curved manifold.

true divergence from a plateau far above it, so we claim the contrast, not divergence itself.) The supercritical plateaus track the law's $e^2/\lambda$ scaling: the implied per-pass excess $e^2(w^*) = \lambda(v_\infty - \sigma^2)$ is 3.9–5.0 $\sigma^2$ across $\lambda \in \{0.25, 0.5, 2/3\}$ (the last from the replay control of Section 6), sitting between the envelope floor and the directly measured per-pass values, as the width-dependence of the measured map requires.

**Scaling to CIFAR-10 pixels (*we observe*).** To test whether the mechanism survives at realistic dimension, we ran the same self-consuming recursion in raw CIFAR-10 pixel space ($32 \times 32 \times 3 = 3{,}072$ dimensions) with a small convolutional denoiser (∼7M parameters, mixed precision, single consumer GPU), $\lambda = \frac{1}{2}$, $G = 8$, three seeds (Figure 7a). Both arms are stable, and they stabilize on opposite sides of the real-vs-real baseline (20.1): the unanchored recursion settles *above* it (off-manifold distance $22.8 \pm 1.2$, a deviation of +2.7, the fattening the theory predicts, same sign in every seed), while the anchored recursion sits *below* it ($16.3 \pm 0.7$, a deviation of −3.8, mild contraction toward the manifold reflecting the anchor's known diversity cost). We are explicit about what panel (a) can and cannot show: the two deviations are of comparable magnitude, so the distance metric alone ranks neither arm; it establishes that the anchored dynamics are stable and non-compounding, not that its samples are closer to the data distribution. The ranking evidence is Figure 7b: against a real-vs-real baseline, the anchored arm keeps *coverage* of the real data at 1.00× (no real modes dropped, ruling out mode collapse) while the unanchored arm loses coverage (1.16×); the anchored arm's reduced intra-set diversity (0.75×) is a bounded cost, identical at generations 4 and 7 (non-compounding), whereas the unanchored arm's coverage loss is the collapse itself. At this model budget the samples are blurry; this is a check on the collapse *dynamics* and a ruling-out of mode collapse, not a demonstration of perceptual quality. The unanchored arm reaches a stable displaced fixed point rather than runaway collapse, exactly as Theorem 1 predicts at $\lambda = \frac{1}{2}$, where fresh-data injection

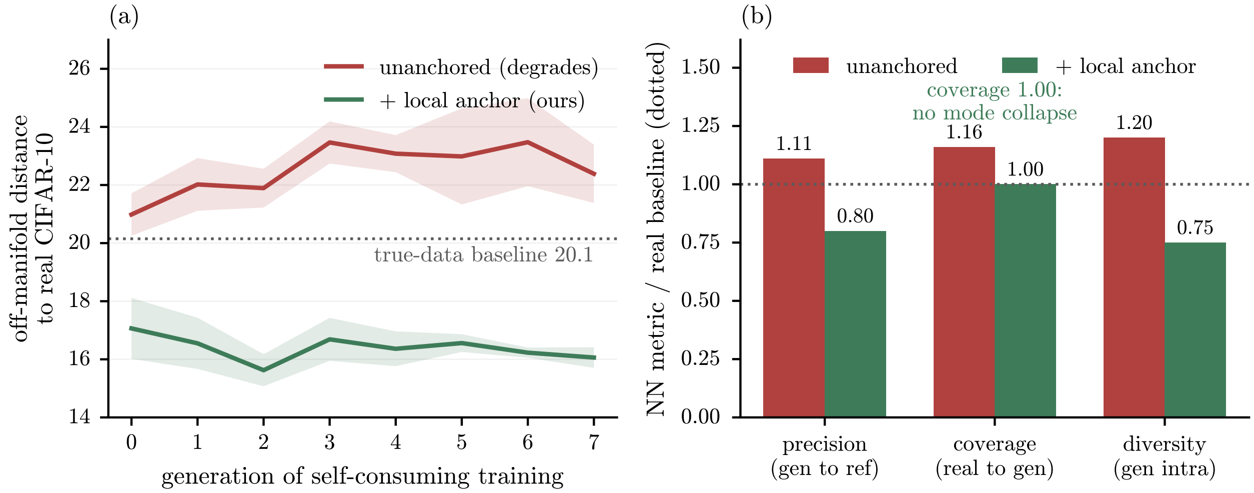


Figure 7: **CIFAR-10 pixel recursion (*we observe*).** (a) Off-manifold distance (mean nearest-neighbor distance to held-out real CIFAR-10) across 8 self-consuming generations, 3 seeds (mean $\pm$ s.d.). The two arms stabilize on opposite sides of the real-vs-real baseline (20.1): unanchored above (+2.7, the predicted fattening), anchored below ($-3.8$, mild contraction toward the manifold); the sign of the split holds in every seed. Because the two deviations are of comparable size, panel (a) alone establishes stability, not superiority; the ranking evidence is (b): three nearest-neighbor metrics relative to a real-vs-real baseline (dotted). The anchor keeps *coverage* of the real data at 1.00$\times$ (no real modes dropped, ruling out mode collapse) while the unanchored arm loses coverage (1.16$\times$); the anchor's 0.75$\times$ diversity is a bounded, non-compounding contraction cost (identical at generations 4 and 7).

pins a finite plateau.

## 8 Limitations and scope

Everything in the *we model* tier is collected here. None of it touches the paper's unconditional core: the closed-form floor (Theorem 2) and the $\Omega(\sqrt{J})$ capacity requirement (Corollary 1) hold given only the slope cap $\bar{\kappa} < \infty$, and the memorylessness theorem (Theorem 3) holds given the frame-cancellation lemma; the items below concern how tightly our *measured* constants pin the recursion's plateau, not whether the floor exists. Item (8) is of a different kind: it delimits which architectures satisfy the slope-cap hypothesis in a binding way, without bearing on the theorem itself. (1) Assumptions 1 and 2 are measured, cross-validated laws about the trained pipeline, not derived facts; we present them as such, in the spirit of measured critical exponents. (2) The recursion floor constant is accounted for empirically, not derived: our closed forms are exact for a single pass and lower bounds for the recursion; feeding the measured capacity degradation $\bar{\kappa}(w)$ back self-consistently closes a quarter to a half (in log) of the once factor-2.4 gap; measuring that degradation under the law (3) constrains, rather than under $p_t$, closes it to 1.1–1.3$\times$ with no further correction, and the per-pass map's one remaining measured channel (in-band profile sag) accounts for the rest (Section 5), at the level of protocol differences. Separately, the *numerical* envelope comparison accounts for 77–85% of the measured floor at $\sigma = 0.05$ and 90–97% at $\sigma = 0.10$ but fails in three of six seeds at $\sigma = 0.13$, where curvature is no longer negligible against the flat-tube idealisation; we take $\sigma \lesssim 0.10$ as the validated range for that comparison. Every step of this accounting is a measurement; none of the

Table 1: Headline numbers (mean ± s.d. over seeds; variance in units of $\sigma^2$ except MNIST and CIFAR-10, which report off-manifold distance).

| Experiment | Arm | Final value | Seeds |
|---|---|---|---|
| Head-to-head, $G$=20 | annealed truncation [1] | $9.84 \pm 1.7$ | 5 |
| | standard sampler + anchor | $1.01 \pm 0.02$ | 5 |
| | jump sampler + anchor | $1.01 \pm 0.02$ | 5 |
| | annealed + replay ($m$=2,000 in-pool) | $8.5 \pm 1.1$ | 5 |
| Critical fraction, $t_0$=$10^{-4}$ | $\lambda = 0.02$ (subcritical) | $69 \pm 16$ at $g_{24}$, rising | 3 |
| | $\lambda = 0.25$ (control) | $16.5 \pm 2.2$ (plateau) | 3 |
| Ablation, fixed $t_0$=$10^{-4}$ | stochastic (noise on) | 10.8 to 13.2 | 3 |
| | deterministic (noise zeroed) | 0 (point collapse) | 3 |
| Pixel MNIST, $G$=8 | unanchored | 3.06 | 3 |
| | anchored | 1.66 | 3 |
| | true-data baseline | 1.34 | n/a |
| Pixel CIFAR-10, $G$=8 | unanchored | $22.8 \pm 1.2$ | 3 |
| | anchored | $16.3 \pm 0.7$ | 3 |
| | true-data baseline | 20.1 | n/a |
| | anchored coverage / diversity | 1.00× / 0.75× | 3 |
| Floor law | envelope at $\bar{\kappa}$ under $p_t$ / under (3) | 3.8 / 5.5 $\sigma^2$ (58% / 81% of measured) | 6 |
| | interventional test, $\bar{\kappa}$: 3.5 → 7.5 | floor 7.6 → {2.5–4.9}, above bound | 3 |

constants ($\bar{\kappa}$, its degradation law — an empirical fit replicated across four synthetic geometries and real MNIST $8 \times 8$ pixel data — or the saturation profile) is derived. (3) The residual-field ($\sim 30\%$ single-pass) channel is bounded in a block-correlation model (Appendix E); negative association across blocks is the one direction that could weaken that particular bound (the closed-form floor of Theorem 2 does not depend on it). (4) Curvature is carried at order $O(s^2/\tau)$ in the flat-tube idealization; sphere experiments bound the resulting bias empirically (non-compounding), but we do not compute the constants. (5) The anchor controls normal *second moments*; tails and tangential statistics improve but are not anchored; on strongly curved data the anchored recursion shows a mild residual creep (Figure 6). (6) A parameter-free one-step prediction of MNIST-latent thin ratios fails quantitatively (it predicts the direction and the regime boundary, not the number); we flag it rather than fit it. (7) Our evidence spans a ring, curved 2-manifolds swept across ambient dimension $d = 8 \to 128$, six $(t_0, \lambda)$ law-identification cells, the two-horned ablation, a replay-buffer control and a subcritical-fraction run, pixel-space MNIST, and a preliminary CIFAR-10 pixel-space recursion (3,072 dimensions, three seeds), all chosen so that every quantity the theory names is directly measurable. The CIFAR-10 run confirms the collapse *dynamics* and rules out mode collapse at scale (Figure 7), but at this model budget the samples are blurry and we do not claim a perceptual-quality result; we do not test latent-diffusion pipelines or larger models. The theory predicts the mechanism transfers, because the floor depends only on the *local* normal slope and not on the ambient dimension (Lemma 4, and the measured $d$-independence of the matched floors); confirming this in a large latent-diffusion system is the clearest next step and is left as future work. (8) **Architectures that factor out the singular normal factor.** The floor's hypothesis is that the learned normal slope is bounded, $\hat{\kappa} \le \bar{\kappa}$, which binds when the network must itself represent the steep normal response. It does *not* bind for parameterizations that build the singular factor into the sampler by construction, for instance a score of the form $-(x - \hat{x})/h(t)$ with a learned projection-like map $\hat{x}$, as in the SiLD construction of Huang et al. [12]: there the network supplies

$\hat{x}$ rather than the slope, and $\hat{\kappa}$ is not limited by network capacity. Theorem 2 is silent about such models. The experiments here all use ordinary $\varepsilon$-prediction, for which the hypothesis does bind. We thank Wei Huang for this observation. The finite-capacity question does not disappear in that setting, it relocates: writing $\hat{x} = \Pi_M(x) + \delta(x)$, the score error carries a term of order $\delta(x)/h(t)$, so projection error is amplified by the same singular factor as $t \to 0$. Whether that amplification admits a floor of the same $\bar{\kappa}^{-2}$ character, with $\mathbb{E}[\delta^2]$ in place of the slope deficit, is open and is the clearest theoretical extension of this work.

# 9 Reproducibility

Code, result logs, and figures are public at

https://github.com/Naman84789/model-collapse-manifolds

Every claim maps to a script, an append-only result log, and a figure in the released code (ring, ablation, replay and critical-fraction controls, MNIST, floor law, law identification; ∼3 CPU-hours total). The result figures regenerate from logged data in seconds. A README lists the run order, the load-bearing scripts, and the headline numbers a reproduction must match. The released code also includes an adversarial verification script for Theorem 2: it re-derives the band crossing symbolically (independent of the manuscript's derivation), integrates the limit ODE with an independent solver, confirms the limiting cases $\rho \to 0$, $\rho \to 1^-$, $\bar{\kappa} \to \infty$, and $\lambda \to 1$, verifies that the boundary layer contracts at the predicted quadratic rate, and attacks the comparison bound with 200 randomized slope profiles including overshooting ones; all checks pass.


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# A Proof of Theorem 1

Write $h(v) := T(v) - v$ with $T$ as in (2) (we reserve $g$ for the generation index); $h$ is concave (composition of an affine map with the concave nondecreasing $w \mapsto w + e^2(w)$, minus a linear term), with $h(0) = T(0) = \lambda\sigma^2 + e^2(\lambda\sigma^2) > 0$ and one-sided derivative $h'(v) = (1-\lambda)\big(1 + e^{2\prime}(w(v))\big) - 1$, nonincreasing in $v$ with limit $(1-\lambda)(1+c_\infty) - 1 < 0$ when $\lambda > \lambda^*$.

**(i) Existence, uniqueness.** A concave function positive at 0 whose derivative is eventually $\leq -\delta < 0$ tends to $-\infty$, so $h$ has a zero; concavity plus $h(0) > 0$ forbids two downcrossings, so the zero $v_\infty$ is unique, with $T(v) > v$ for $v < v_\infty$ and $T(v) < v$ for $v > v_\infty$.

**Contraction above.** For $v \geq v_\infty$, monotonicity and concavity give $0 \leq T(v) - v_\infty \leq \rho_{\mathrm{loc}}(v - v_\infty)$ with $\rho_{\mathrm{loc}} = T'(v_\infty^+) < 1$. The strict inequality needs no extra hypothesis: if $\rho_{\mathrm{loc}} = 1$ then $h'(v_\infty) = 0$, making the concave $h$ maximal at $v_\infty$, whence $h(v_\infty) \geq h(0) = T(0) > 0$, contradicting $h(v_\infty) = 0$. So the crossing is always strict and the rate genuinely geometric.

**Convergence under vanishing perturbations.** Let $v_g = T(v_{g-1}) + s_g^2$, $s_g^2 \to 0$. While iterates stay $\geq v_\infty$, $x_g := v_g - v_\infty \leq \rho_{\mathrm{loc}} x_{g-1} + s_g^2 \to 0$ by the standard perturbed contraction argument (split $\sum_j \rho_{\mathrm{loc}}^{g-j} s_j^2$ at $j = g/2$). If an iterate drops below $v_\infty$: define the unperturbed comparison $u_g = T(u_{g-1})$ started there; then $u_g \uparrow v_\infty$ (monotone, bounded, $T(v) > v$ below $v_\infty$) and $v_g \geq u_g$ by induction ($s_g^2 \geq 0$, $T$ monotone), so $\liminf v_g \geq v_\infty$, while $v_g < v_\infty + s_g^2$ gives $\limsup v_g \leq v_\infty$. Alternating regimes inherit both barriers. Linearizing $T$ at $v_\infty$ gives the local rate.

**(ii) Divergence.** If $(1-\lambda)(1+c_\infty) > 1$ then, since $e^{2\prime} \geq c_\infty$ everywhere (the concave derivative decreases *to* its limit), $h' \geq (1-\lambda)(1+c_\infty) - 1 > 0$, so $h$ is nondecreasing and $h(v) \geq h(0) > 0$ for all $v$: after $G$ generations $v_G \geq v_0 + G\,T(0) \to \infty$, regardless of the schedule ($s_g^2 \geq 0$ only helps).

**(iii)** Substituting $e^2(w) = e_0^2 + cw$ makes $T$ affine with slope $(1+c)(1-\lambda)$; solve the fixed point. □

# B The variance ODE

**Lemma 1.** One Euler step of the reverse scheme is $x' = x + \big(\frac{1}{2}x - \hat{\varepsilon}(x,t)/s\big)dt + \sqrt{dt}\,\xi$ with $\xi$ independent of $x$. Write $f(x) = \frac{1}{2}x - \hat{\varepsilon}(x,t)/s$. By bilinearity of the covariance and independence of the injected noise,

$$\mathrm{Var}(x') = \mathrm{Var}(x) + 2\,dt\,\mathrm{Cov}\big(x, f(x)\big) + dt^2\mathrm{Var}\big(f(x)\big) + dt = V + \Big[V - \tfrac{2}{s}\mathrm{Cov}\big(x, \hat{\varepsilon}(x)\big) + 1\Big]dt + O(dt^2).$$

No property of $\hat{\varepsilon}$ beyond square-integrability is used, so this holds for a nonlinear estimator; the $+dt$ is the injected noise and the $O(dt^2)$ term is the finite variance of the drift. Divide by $dt$, let $dt \to 0$, and substitute $\mathrm{Cov}(x, \hat{\varepsilon}(x)) = \hat{\kappa}\mathrm{Var}(x)$ from (3) to get (4).

*Independent route, no discretisation.* For the continuous reverse SDE $dX = f(X)d\tau + dW$, Itô gives $d(X^2) = 2Xf(X)d\tau + 2X\,dW + d\tau$, so $\mathbb{E}[X^2]' = 2\mathbb{E}[Xf(X)] + 1$ while $\mathbb{E}[X]' = \mathbb{E}[f(X)]$; subtracting, $\mathrm{Var}' = 2\mathrm{Cov}\big(X, f(X)\big) + 1$, which is (4). With $\hat{\kappa} = \kappa = s/\gamma^2$: $\kappa/s = 1/\gamma^2$ and $V = \gamma^2$ solves the equation exactly (direct check), confirming the bookkeeping. An intercept in $\hat{\varepsilon}$ moves only the mean. □

**Remark 2.** For an i.i.d. copy $X'$ of $X$, $\frac{1}{2}\mathbb{E}[(X - X')(g(X) - g(X'))]$ expands to $\mathbb{E}[Xg(X)] - \mathbb{E}[X]\mathbb{E}[g(X)] = \mathrm{Cov}(X, g(X))$. If $g$ is $\bar{\kappa}$-Lipschitz then $(X - X')(g(X) - g(X')) \leq \bar{\kappa}(X - X')^2$ pointwise, and $\mathbb{E}[(X - X')^2] = 2\mathrm{Var}(X)$, so $\mathrm{Cov}(X, g(X)) \leq \bar{\kappa}\mathrm{Var}(X)$ under any law. □

## C Floor lemmas and closed forms

**Lemma 2.** From (1), $\kappa(t) = s/(a^2\sigma^2 + s^2)$ with $s^2 = 1 - e^{-t}$: $\kappa \to 0$ as $t \to 0$ ($s \to 0$, $\gamma^2 \to \sigma^2$), $\kappa \approx 1/s$ for $s^2 \gg \sigma^2$, and maximizing over $s$ gives $\kappa_{\max} = 1/(2\sigma)$ at $s = \sigma$ (i.e. $t \approx \sigma^2$), to leading order in $\sigma^2$. Below the band the remaining e-folds are $\int_{t_0}^{t_-} 2\kappa/s\,dt = \int 2\,dt/\gamma^2 = 2\log\frac{\sigma^2+t_-}{\sigma^2+t_0} \leq 2\log(1+\rho^2/4) \leq \rho^2/2$, using $t_- \approx \bar{\kappa}^2\sigma^4$. □

**Lemma 3.** Substitute $t = \sigma^2 u$, $V = \sigma^2 W$: $s^2 = \sigma^2 u(1 + O(\sigma^2 u))$, $\gamma^2 = \sigma^2(1+u)(1+O(\sigma^2 u))$, so $\kappa = (1/\sigma)\frac{\sqrt{u}}{1+u}(1 + O(\sigma^2 u))$ and $2\hat{\kappa}/s = (2m(u)/\sqrt{u})/\sigma^2$; the ODE becomes $-dW/du = 1 - (2m/\sqrt{u})W + \sigma^2 W(1 + o(1))$. On $u \in [0, U]$ the perturbing terms are $O(\sigma^2 U)$ *per unit of* $W$; the band occupies $u \leq u_+ = O(1/\rho^2)$, giving the stated *relative* error $O(\sigma^2/\rho^2)$ (the additive gap to $g$ is a factor $W \sim \rho^{-2}$ larger, hence $O(\sigma^2/\rho^4)$). For the boundary: the uncapped equation $-dW/du = 1 - 2W/(1+u)$ has general solution $W = (1+u) + A(1+u)^2$ (direct check), so deviations from the slave solution $W = 1 + u$ contract as $\big((1+u)/(1+u')\big)^2$ under downward integration from $u'$ to $u$; the influence of the $t \gtrsim 1$ boundary layer at band entry is suppressed by this quadratic factor and is absorbed in the same error term. *Numerical check:* the slave solution is invariant to $3 \times 10^{-15}$ under the integrator; finite-$\sigma$ floors converge to the limit values from above (e.g. $\rho = 0.39$: 3.953/3.812/3.775 at $\sigma = 0.10/0.05/0.02$ vs. limit 3.769). □

**Theorem 2(U).** By the scalar comparison principle (equal forcing and data; smaller $\hat{\kappa}$ means larger $V$), any profile with $\hat{\kappa} \leq \bar{\kappa}$ has $V \geq V_{\hat{\kappa}\equiv\bar{\kappa}}$, one-sided; no lower bound on $\hat{\kappa}$ is used. For the constant envelope, (5) with $m \equiv \rho/2$ has integrating factor $e^{-2\rho\sqrt{u}}$: $\frac{d}{du}\big[We^{-2\rho\sqrt{u}}\big] = -e^{-2\rho\sqrt{u}}$, so for data of sub-quadratic growth at $u = \infty$, $W(0^+) = \int_0^\infty e^{-2\rho\sqrt{u}}du = 2\int_0^\infty xe^{-2\rho x}dx = \frac{1}{2\rho^2}$, exactly, for every $\rho > 0$. $1/(2\rho^2) > 1 \iff \rho < 1/\sqrt{2}$.

**Theorem 2(N).** The band edges solve $\sqrt{u}/(1+u) = \rho/2$, a quadratic in $x = \sqrt{u}$: $\rho x^2 - 2x + \rho = 0$, $x_\mp = (1 \mp q)/\rho$, $q = \sqrt{1-\rho^2}$ (nonempty iff $\rho < 1$). Above the band the solution is the exact slave $W = 1 + u$; it enters the band with $W(u_+) = 1 + u_+$. Across the band ($m = \rho/2$), the same integrating factor with antiderivative $F(u) = -(2\rho\sqrt{u} + 1)e^{-2\rho\sqrt{u}}/(2\rho^2)$ gives

$$W(u_-) = (1+u_+)e^{-2\rho(x_+-x_-)} + \frac{(2\rho x_- + 1) - (2\rho x_+ + 1)e^{-2\rho(x_+-x_-)}}{2\rho^2},$$

and $2\rho x_\mp = 2(1 \mp q)$, $2\rho(x_+ - x_-) = 4q$ yield the displayed $W_{\text{exit}}$. Below the band the equation is uncapped with exact general solution $(1+u) + A(1+u)^2$; matching at $u_-$ gives $A = \big(W(u_-) - (1 + u_-)\big)/(1+u_-)^2$ and $g(\rho) = W(0^+) = 1 + A$, which is (7). Limits: as $\rho \to 1^-$ ($q \to 0$), $u_\pm \to 1$ and $g \to 1$; as $\rho \to 0$ ($q \to 1$), $u_+ \approx 4/\rho^2$, $u_- \approx \rho^2/4$, and expanding, $g = \big[8e^{-4} + 1 - 5e^{-4}\big]/(2\rho^2) + O(1) = (1 + 3e^{-4})/(2\rho^2) + O(1)$, giving $C_0 = (1 + 3e^{-4})/2$. Positivity $g > 1$ on $(0,1)$ follows from $W(u_-) > 1 + u_-$ (the band under-contracts relative to the slave). *Numerical check:* (7) and the exact $1/(2\rho^2)$ match direct integration of (5) to $\leq 2.4 \times 10^{-5}$ over $\rho \in [0.1, 1]$ (pure Euler residual).

Both bounds apply to the fully nonlinear estimator with no extension step, since (4) is exact and $\hat{\kappa}$ is defined under the sampler's own law (Lemma 1). Lemma 3 transfers the limit values to finite $\sigma$ with the stated error. □

**Corollary 1** is Theorem 2(U) read at fixed $\bar{\kappa}$ with $\sigma \to 0$; the capacity bound inverts $1/(2\rho^2) \leq C$. □

# D Posterior-mean sharpening and per-axis necessity

**Proposition 1** (Tweedie sharpening; *we prove*)**.** *In the Gaussian normal coordinate with axis variance $\sigma^2_{\mathrm{ax}}$, the jump with the exact predictor outputs the posterior mean, with*

$$\mathrm{Var}(\hat{x}_0) = \frac{a^2\sigma^4_{\mathrm{ax}}}{\gamma^2_{\mathrm{ax}}(\hat{t})}, \qquad \frac{\mathrm{Var}(\hat{x}_0)}{\sigma^2_{\mathrm{ax}}} = \frac{a^2\sigma^2_{\mathrm{ax}}}{a^2\sigma^2_{\mathrm{ax}} + s^2(\hat{t})} < 1,$$

*and with a trained predictor $\hat{\varepsilon} = \varepsilon^* - \tilde{\delta}$, $v_{\mathrm{jump,ax}} = a^2\sigma^4_{\mathrm{ax}}/\gamma^2_{\mathrm{ax}} + (s^2/a^2)\mathbb{E}[\tilde{\delta}^2] + 2(s/a)\,\mathrm{Cov} + O(s^4/\tau^2)$.*

*Proof.* $\hat{x}_0 = (x - s\varepsilon^*)/a = x(1 - s^2/\gamma^2)/a = (a\sigma^2_{\mathrm{ax}}/\gamma^2)x$, so $\mathrm{Var} = a^2\sigma^4_{\mathrm{ax}}/\gamma^2$; Tweedie's formula identifies this with $\mathbb{E}[X_0 \,|\, X_{\hat{t}}]$ [14]. The error term is the jump map applied to $\varepsilon^* - \tilde{\delta}$; the curvature term is the posterior-mean chord bias. □

Consequences: strongly singular axes ($\sigma_{\mathrm{ax}} \ll s$) are collapsed onto the manifold (the matcher must top up); fat axes ($\sigma_{\mathrm{ax}} \gg s$) are untouched while injection fattens them (the matcher must contract); axes with $\sigma_{\mathrm{ax}} \sim s$ sharpen by an axis-dependent factor strictly inside $(0, 1)$. Hence the required correction is signed and anisotropic: any correct matcher must be bidirectional and per-axis. (*We observe:* scalar matching leaves thin ratios at 0.62 to 0.65 on MNIST latents; per-axis matching achieves $|\mathrm{thin} - 1| \leq 0.012$.)

# E The residual channel (*we model*)

(*Role revised by measurement*: the floor anatomy of Section 5 identifies the single-pass remainder beyond the envelope as one measured channel — in-band profile sag, the ensemble-displacement effect having been absorbed into the hypothesis by (3) — with the residual beyond it small and slightly negative; this appendix's stochastic bound is retained as a consistency check on the residual field's magnitude, not as the remainder's explanation.)

The residual of the estimator about its affine $L^2(p_t)$ fit, $\delta := \hat{\varepsilon} - \hat{\kappa}_{\mathrm{aff}}x - b$ (used in this appendix only, as an independent characterisation of the residual field; the floor proof does not rely on it), has measured band risk $r^2_{\mathrm{res}}(t) \approx 0.02$ to $0.05$, flat in $t$ and in sample size, with path-coherence time $\approx \nu t$, $\nu \approx 0.75$. Under a block-correlation model (coherent within blocks of length $\nu t$, independent across blocks, no systematic over-contraction), the injected variance obeys $\Phi_0(t_0) \geq c\,\nu\rho_0(\sigma^2 + t_0)$: decreasing in $t_0$ until $t_0 \sim \sigma^2$, then saturating, the same shape as the measured excess (knee at $t_0 \approx \sigma^2$, $n$-flat), with the single constant effective to within a factor $\sim 2$. In this paper the channel plays a supporting role only: it is the measured $\sim 30\%$ single-pass remainder on top of the closed-form floor, which does not depend on it.

# F Experimental details

**Networks and training.** 3-layer MLPs, width 256, SiLU, 32-dim sinusoidal time embedding; Adam, learning rate $2 \times 10^{-3}$; batch 512 (ring) / 256 (MNIST); denoising score matching with $t \sim U(10^{-3}, 8)$; fresh network each generation.

**Sampling.** Reverse Euler-Maruyama on a 200-point geometric grid from $T = 8$; the ablation zeroes the $\sqrt{|dt|}\,\xi$ term only. Jump: $\hat{t} = 0.05$.

**Anchor.** Ring: radial (normal) bidirectional match of the pool's radial variance to the stale reference's, contract-by-scaling / top-up-by-noise. MNIST: per-point $k$NN-PCA ($k = 96$, $k_{\mathrm{dim}} = 12$) charts from the reference; normal coordinates rescaled to the reference's local normal eigenvalues in the same frames. Reference: $m = 2{,}000$ real samples drawn once at $g = 0$, never refreshed (provenance-free: the matcher is applied to the whole pool without labels).

**Measured network profile.** The fitted slope $\hat{\kappa}(t)$ is extracted by affine regression of $\hat{\varepsilon}$ against $x$ on fiber lines at each $t$; it tracks $\kappa(t)$ (0.95 to 0.97 for $t \geq 0.1$), saturates at $\bar{\kappa} \approx 3.9$ inside the band, and never exceeds $\kappa$ ($n$-independent across 1k to 16k) — a data-width statement: nets trained on fat pools ($\sqrt{w} \gtrsim 0.10$) show mild overshoot at mid times, so the class-(N) constant applies at the data width while class (U) needs no such assumption. The decomposition identity $(\kappa - \hat{\kappa})^2\gamma^2 + r_{\mathrm{res}}^2 = r_{\mathrm{tot}}^2$ checks numerically along the path.

**$t_0$-dependence of the response slope.** The probe's net slope $c \in \{0.00, \ldots, 0.13\}$ is measured at its sampling point $t_0 = 5 \times 10^{-3}$. At the recursion's operating point $t_0 = 10^{-4}$, the directly measured per-pass map over the same widths has $dv/dw \approx 0.68$ ($c \approx -0.32$): once the sampler traverses the whole band, the deficit channel dominates and $e^2$ *decreases* over this range — still within the law's requirement ($e^{2\prime} > -1$ keeps $T$ monotone and concave, and only strengthens contraction), and consistent with the plateau measurements ($e^2(w^*) = 3.9$–$5.0\,\sigma^2$ falling with the fixed point's width). The $\lambda = 0.02$ runaway then implies the slope turns back positive in the far-fat regime ($w \gg 10\,\sigma^2$), outside the anatomy's measured range; Assumption 2's global nondecreasing-concave form is a regime model, exact where identified.

**Anatomy of $c$.** Training single-width tubes at $w \in \{0.0009, \ldots, 0.0169\}$ and measuring $dv/dw - 1$ gives net $c \in \{0.00, 0.05, 0.13, 0.10, 0.05\}$ (small, positive, concave) as the sum of a deficit response $\approx -0.68$ (a fatter pool needs a smaller peak slope) and an injection response $\approx +0.73$ (a fatter pool damps injected variance less). The same probe's per-width slope fits give the capacity-degradation fit $\bar{\kappa}(w) = 4.43 - 11.6\sqrt{w}$ used in Section 5, which refines this anatomy into three channels: writing $\frac{d}{dw}\Phi_{\mathrm{det}}(w, \bar{\kappa}(w)) = \partial_w \Phi_{\mathrm{det}}|_{\bar{\kappa}} + (\partial_{\bar{\kappa}}\Phi_{\mathrm{det}})\,\bar{\kappa}'(w)$, the mid-range values are a deficit $\approx -0.94$ and a capacity-degradation term $\approx +0.7$ (the latter is not a fit artifact: raw centered differences of the five measured points, with no functional form at all, give $+0.78/+0.68/+0.67$ at the three interior widths vs. the fit's $+0.84/+0.70/+0.65$). The derived deterministic response is therefore $\approx -0.23$ mid-range, and the measured total ($+0.10$ to $+0.13$) leaves a residual attributed to injection of $\approx +0.3$ to $+0.4$. Because injection is *defined* as this residual, the three channels sum to the measurement by construction; the non-trivial content is that the residual comes out stable, positive, and of the size the residual-channel model (Appendix E) expects, and that most of what the two-channel anatomy above attributes to injection is really $\bar{\kappa}$ degrading on fatter pools. We trust the split mid-range only: at the thinnest width $\bar{\kappa}'(w)$ from the $\sqrt{w}$ fit diverges, an artifact of the parametrization, not a mechanism.